1. Clean State

The system is reset by removing previous containers, logs, and unloading the kernel module to ensure a fresh environment.

```
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$ cd ~/OS-Jackfruit/boilerplate

sudo pkill engine
sudo rmmod monitor
sudo rm -rf /tmp/container_logs
sudo rm -f /tmp/engine_supervisor.sock
[sudo] password for shri:
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$
```

2. Load Module

The kernel module is loaded using insmod. The system confirms successful loading and creates the device `/dev/container_monitor`.

```
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo insmod monitor.ko
sudo dmesg | grep container_monitor
[ 4085.857935] container_monitor: unloaded, all entries freed
[ 4085.945524] container_monitor: loaded (major=239)
[ 4085.945531] container_monitor: create device with:
[ 4085.945532]   sudo mknod /dev/container_monitor c 239 0
[ 4085.945533]   sudo chmod 666 /dev/container_monitor
[ 4332.889596] container_monitor: registered 'test1' pid=11428 soft=4096KB hard=8192KB
[ 4334.255907] container_monitor: container 'test1' (pid=11428) no longer exists, removing
[ 4652.701925] container_monitor: registered 'test2' pid=11747 soft=4096KB hard=8192KB
[ 4654.527729] container_monitor: 'test2' (pid=11747) RSS=2636 KB (soft=4096, hard=8192)
[ 4656.554944] container_monitor: [SOFT LIMIT] container 'test2' (pid=11747) using 4684 KB, soft limit is 4096 KB
[ 4656.554955] container_monitor: 'test2' (pid=11747) RSS=4684 KB (soft=4096, hard=8192)
[ 4658.607001] container_monitor: 'test2' (pid=11747) RSS=6732 KB (soft=4096, hard=8192)
[ 4660.647948] container_monitor: [HARD LIMIT] container 'test2' (pid=11747) using 8780 KB, hard limit is 8192 KB — kill
ing
[ 4660.647972] container_monitor: [HARD LIMIT] killed container 'test2' (pid=11747)
[ 5516.601928] container_monitor: unloaded, all entries freed
[ 5560.063995] container_monitor: loaded (major=239)
[ 5560.064001] container_monitor: create device with:
[ 5560.064002]   sudo mknod /dev/container_monitor c 239 0
[ 5560.064003]   sudo chmod 666 /dev/container_monitor
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$
```

3. Start Supervisor

The engine supervisor is started, which manages container lifecycle and communicates with the kernel module.

```
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$ ls /dev/container_monitor
/dev/container_monitor
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine supervisor ./rootfs-base
[supervisor] ready on /tmp/engine_supervisor.sock
```

4. Run Memory Test

A test container is started using memory_hog, which continuously allocates memory to simulate increasing usage.

```
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start test ./rootfs-alpha /memory_hog --soft-mib 4 --hard
-mib 8
[sudo] password for shri:
OK: started 'test' (pid=13341)
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$
```

5. Soft Limit & Hard Limit

When memory usage crosses the soft limit (4MB), a warning is logged.
When it exceeds the hard limit (8MB), the kernel module terminates the process.

```
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo dmesg -C
[sudo] password for shri:
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo dmesg -w
[ 6080.241007] audit: type=1400 audit(1776925839.417:319): apparmor="DENIED" operation="open" class="file" profile="snap
.firmware-updater.firmware-notifier" name="/proc/sys/vm/max_map_count" pid=13133 comm="firmware-notifi" requested_mask="
r" denied_mask="r" fsuid=1000 ouid=0
[ 6224.070630] container_monitor: registered 'test' pid=13341 soft=4096KB hard=8192KB
[ 6225.041128] container_monitor: 'test' (pid=13341) RSS=1592 KB (soft=4096, hard=8192)
[ 6227.083204] container_monitor: 'test' (pid=13341) RSS=3640 KB (soft=4096, hard=8192)
[ 6229.134425] container_monitor: [SOFT LIMIT] container 'test' (pid=13341) using 5688 KB, soft limit is 4096 KB
[ 6229.134432] container_monitor: 'test' (pid=13341) RSS=5688 KB (soft=4096, hard=8192)
[ 6231.174843] container_monitor: [HARD LIMIT] container 'test' (pid=13341) using 8760 KB, hard limit is 8192 KB — killi
ng
[ 6231.175050] container_monitor: [HARD LIMIT] killed container 'test' (pid=13341)
```

6. Check Status

The container status shows “killed”, confirming enforcement of memory limits.

```
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start test ./rootfs-alpha /memory_hog --soft-mib 4 --hard
-mib 8
[sudo] password for shri:
OK: started 'test' (pid=13341)
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
NAME         PID      STATE        UPTIME(s)  SOFT(MiB) HARD(MiB)
-------------------------------------------------------------------
test         0        killed       146        4         8
shri@shri-VirtualBox:~/OS-Jackfruit/boilerplate$
```

The system successfully enforces memory limits using the kernel module. Soft limit generates warnings, and hard limit ensures process termination, validating correct implementation.